

Lecture #12

Content to be covered

- ❑ Production of polarized light
 - ❖ By double refraction
 - ❖ By selective absorption
- ❑ Birefringent Materials
- ❑ Nicole Prism

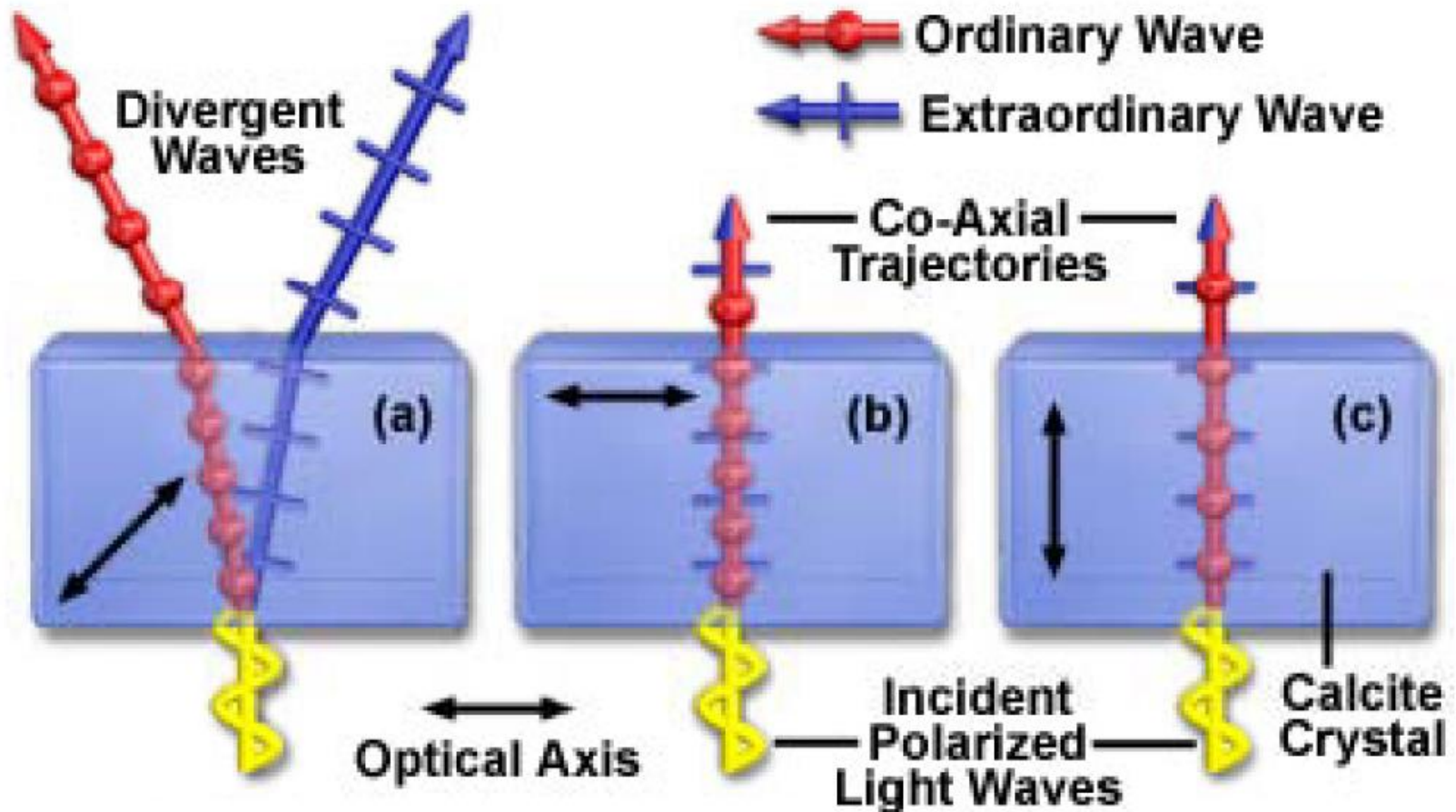
Resources to be consulted

- ❑ OPTICS by Ajay Ghatak
- ❑ <https://micro.magnet.fsu.edu/primer/java/polarizedlight/nicol/index.html>

Double Refraction

- ❖ Many transparent solids are optically isotropic, meaning that the index of refraction is equal in all directions through out the crystalline lattice. Examples of isotropic solids are glass, table salt (sodium chloride, many polymers, and a wide variety of both organic and inorganic compounds.
- ❖ Anisotropic crystals, on the other hand, have crystallographically distinct axes and interact with light in a manner that is dependent upon the orientation of the crystalline lattice with respect to the incident light.
 - When light enters along the **optical axis** of anisotropic crystals, it acts in a manner similar to interaction with isotropic crystals and passes through at a single velocity.
 - However, when light enters along any non-equivalent axis, it is refracted into two rays each polarized with the vibration directions oriented at right angles to one another, and traveling at different velocities. This phenomenon is termed "double" or "bi" refraction and is seen to a greater or lesser degree in all anisotropic crystals.

Separation of light by a Birefringent Materials



both rays travel
with different velocities
in different directions

both rays travel
with different velocities
in same direction

both rays travel
with same velocity
in same direction

Theory of Double Refraction in Uniaxial Crystal

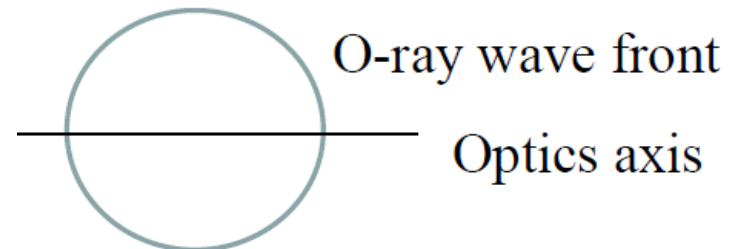
To explain the double refraction following experimental observation must be considered:

- ❑ Ordinary rays (o-ray) have same velocity in all direction.
- ❑ Extraordinary rays (e-ray) *have different velocity in* different direction inside the crystal.
- ❑ along optic axis velocities of the two rays are same.
- ❑ both rays travel along the same path but with different velocities in a direction perpendicular to the optic axis.

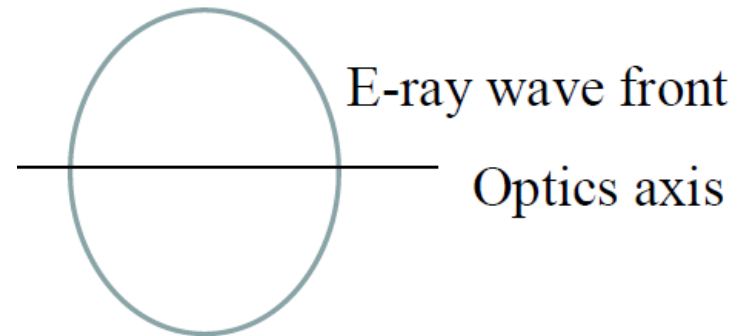
Huygen's Theory Of Double Refraction In Uniaxial Crystal

According to wave theory of light (using the principle of secondary wavelets):
When the light wave strikes the surface of a doubly refracting crystal, every point of the crystal distributed by the incident wavefront becomes the source of two secondary wavelets, ordinary and extraordinary, which spreads out into the crystal.

For O-ray the velocity is same in all directions,
the wavefront is spherical.

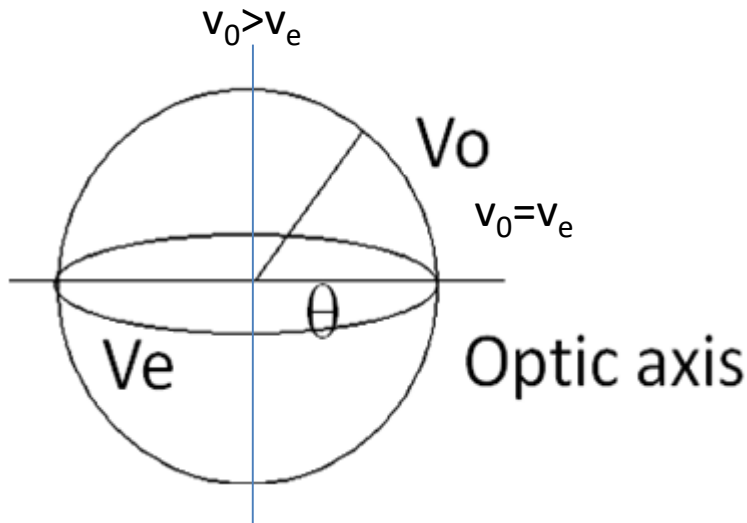


For E-ray the velocity is different in different
directions the wavefront is an ellipsoid.



Positive Uniaxial Crystal

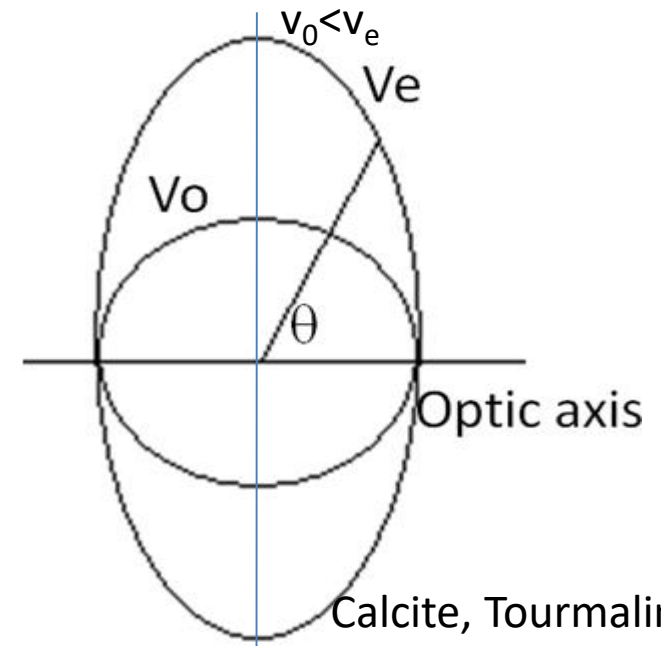
In **quartz** the velocity of O ray is greater than velocity of E ray *i.e.* $v_o > v_e$ so $\mu_o < \mu_e$ and $r_o > r_e$



Quartz
+ve crystal

Negative Uniaxial Crystal

In **calcite** the velocity of O ray is less than velocity of E ray *i.e.* $v_o < v_e$ so $\mu_o > \mu_e$ and $r_o < r_e$



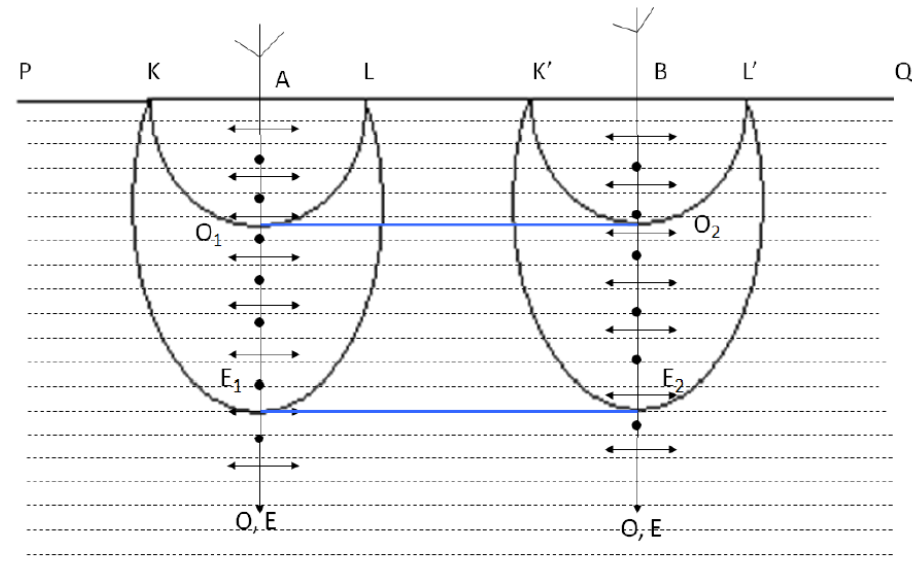
Calcite, Tourmaline
-ve crystal

Along the optic axis $v_o = v_e$, both wave front will touch each other

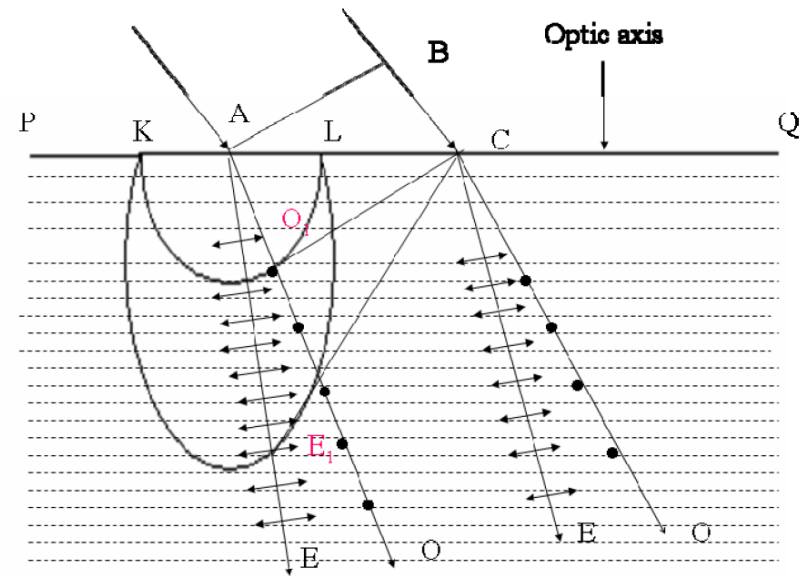
In negative uniaxial crystals the sphere lies inside the ellipsoid, while in positive uniaxial the ellipsoid lies inside the sphere.

Optic axis parallel to the refracting surface and in the plane of incidence

Normal Incidence



Oblique Incidence



Polarization by Double Refraction

The “double bending” of the beam transmitted through crystals is called **Double refraction or Birefringence**.

Crystals displaying this property are called **Birefringent crystals**. Example: **Calcite, Tourmaline, Quartz, etc.**

When an un-polarized light enters into an anisotropic crystal, it splits into two beams, each of them being characterized by a certain state of polarization. If we could eliminate one of the beams by some method, then we would obtained a polarized light.

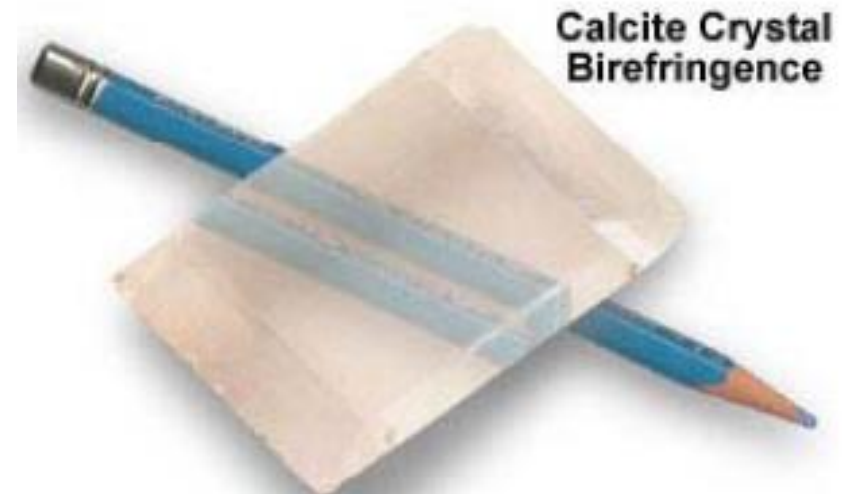
Methods of eliminating one of the beam.

1. Selective absorption - Dichroism.

Example: Tourmaline crystal (Dichroic Crystal)

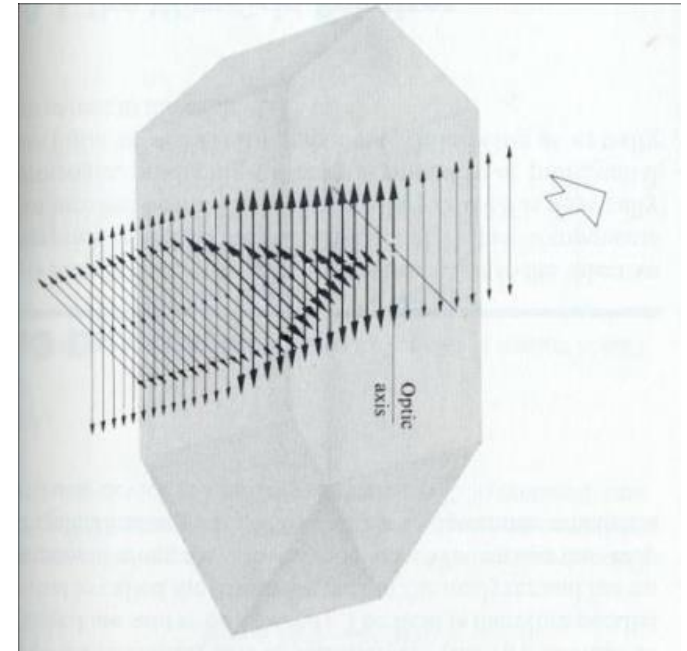
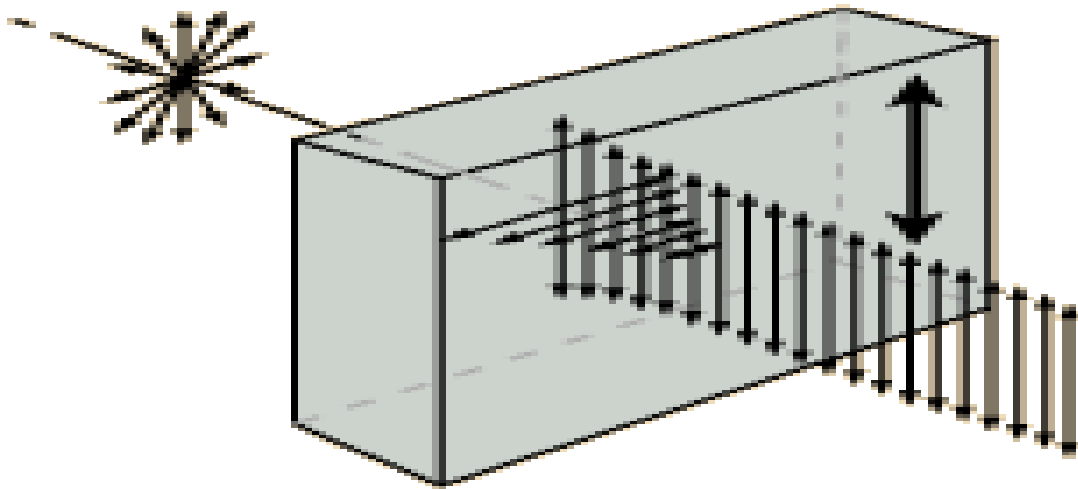
2. Total Internal Reflection

Example: Nicole prism



Polarization by Absorption: Dichroic materials

A number of crystalline materials absorb more light in one incident plane than another, so that light progressing through the material become more and more polarized as they proceed. This **anisotropy in absorption is called dichroism**. There are several naturally occurring [dichroic materials](#), and the commercial material [polaroid](#) also polarizes light by **selective absorption** technique.



Tourmaline crystal is a dichroic material

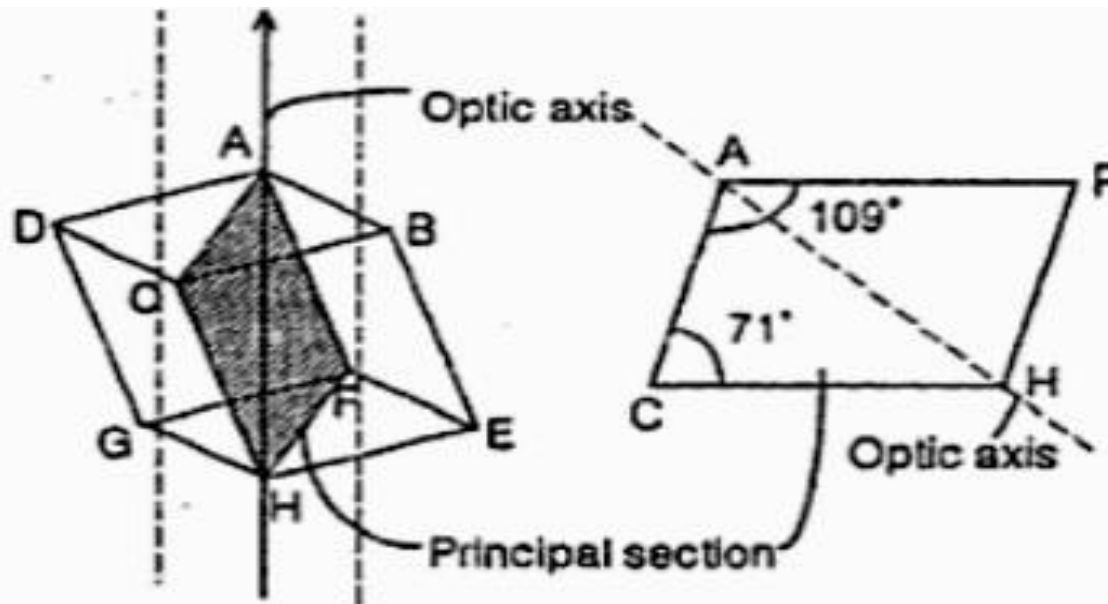
Principal Planes and Principal Sections

Principal planes:

The plane containing optic axis. Many such planes can be drawn.

Principal Section:

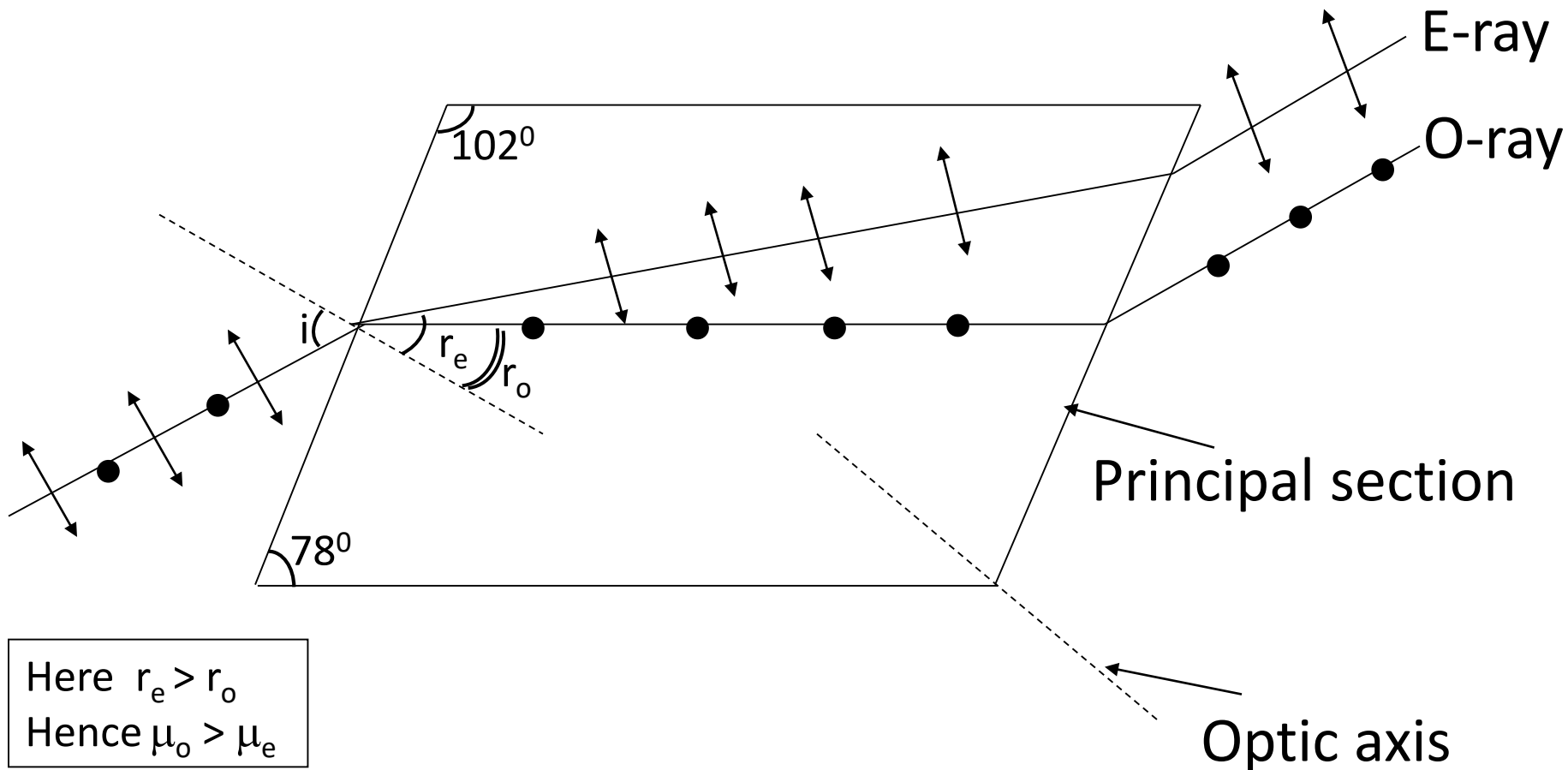
The principal plane normal to a pair of opposite refracting faces is called as **principal section** for that pair of faces. Since there are six faces, therefore there are **three principal sections** in a crystal corresponding to each pair of opposite faces.



Polarization by Double Refraction

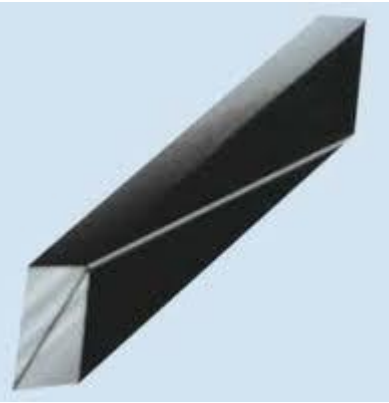
When un-polarized light passes through a uniaxial crystal it splits up into two refracted rays.

- Ordinary wave(O-wave)
- Extra-ordinary wave(E-wave)



Nicol Prism

Removal of one of the two refracted beams in case of doubly refracting calcite crystal By **total internal reflection**.



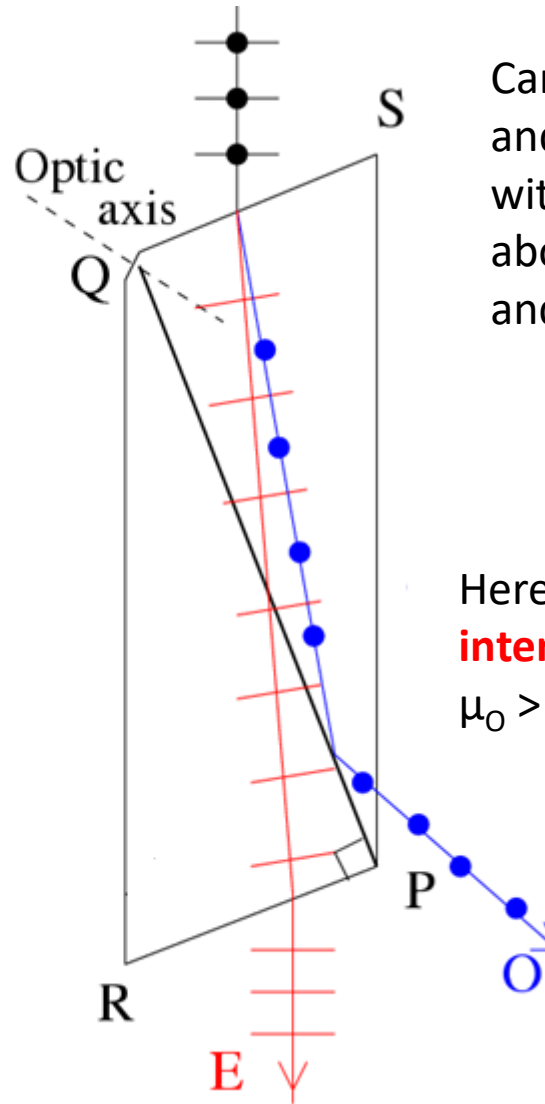
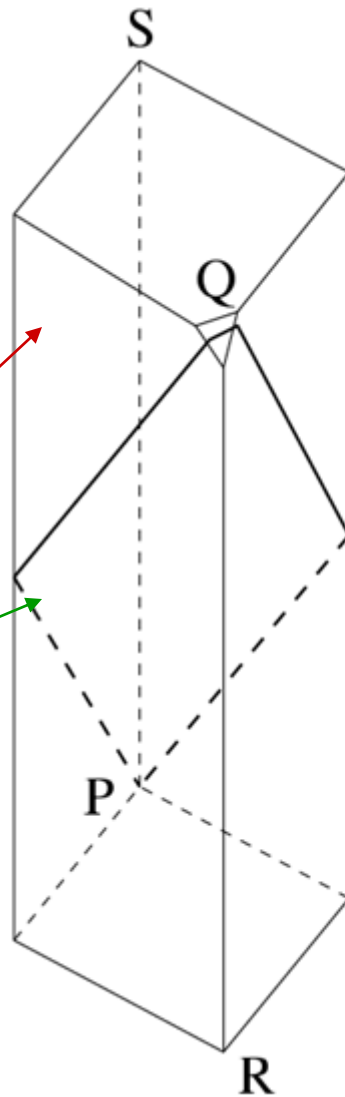
Calcite

$$\mu_o = 1.6584$$

$$\mu_e = 1.4864$$

Canada balsam

$$\mu_{CB} = 1.55$$



Canada Balsam is a clear and transparent substance with an index of refraction about midway between μ_o and μ_e i.e. $\mu_o > \mu_{CB} > \mu_e$

Here, **O-ray will have total internal reflection** because $\mu_o > \mu_{CB}$.

Double Refraction : Summary

- Along optic axis velocities of the two rays are same as the double refraction is absent. Sum of the intensities of the two rays equal to the intensity of the incident ray.
- Both rays travel along the same path but with different velocities in a direction perpendicular to the optic axis.
- The ordinary ray is plane polarized in the principal plane of the crystal, *i.e.* vibrations of the ordinary wave are perpendicular to the principal plane of the crystal.
- Extraordinary ray is also plane polarized in the plane at right angle to the principal plane, means it's vibrations are in the principal plane.
- Difference between the refractive indices for O-ray and E-ray is known as **birefringence** = $(\mu_o \sim \mu_e)$.